The Induction Principle — problem 8.24

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Let $a_1, \ldots, a_n$ be positive integers such that $a_1 \leq \cdots \leq a_n$. Prove that

$$\frac{1}{a_1} + \cdots + \frac{1}{a_n} = 1 \Rightarrow a_n < 2^{n!}.$$